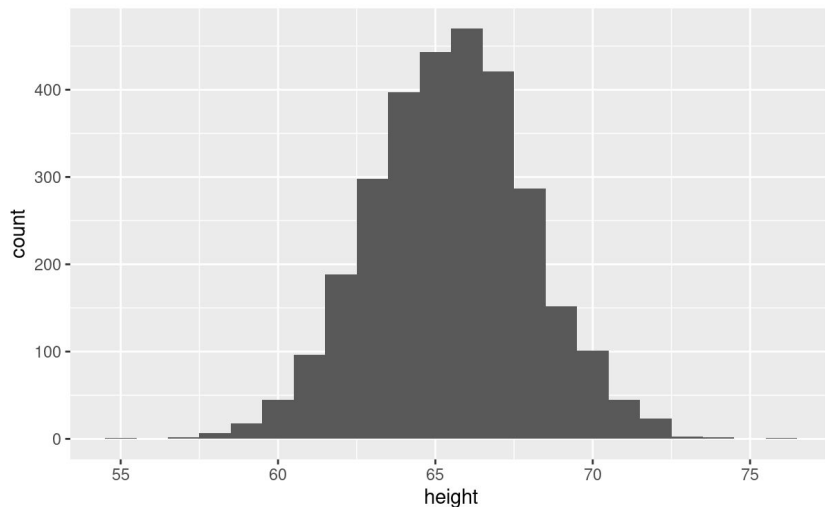


Intro to t-distribution

MATH 213

Group 1

Put your graph showing the distribution of heights in the population here.

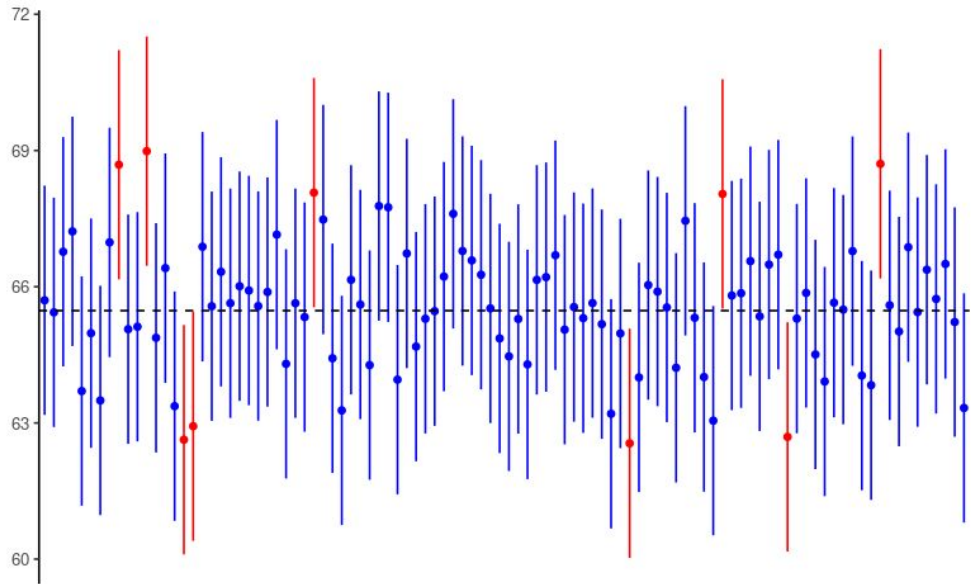


Describe the population distribution of heights, following our standard procedure (center, spread, shape):

The graph describes the heights of prisoners in England and Wales. The graph is a normal bell curve, has a spread of 55 in to 76 in, and a mean height of 66 inches tall. Most of the prisoner's heights between 63.73 and 67.16 inches tall

How many of the 100 95% confidence intervals (calculated using the actual standard error) do you predict will contain the population mean height? 95

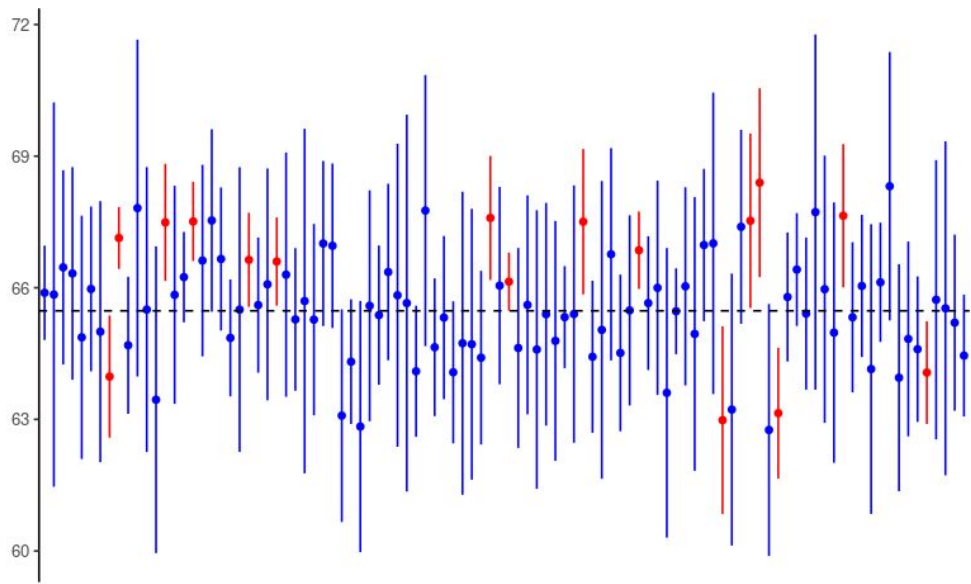
Put your graph of the 100 confidence intervals for your 100 samples here.



How many of your 100 confidence intervals actually contained the population mean? 91

How many of the 100 95% confidence intervals (calculated using the **estimated** standard error) do you predict will contain the population mean height? 95

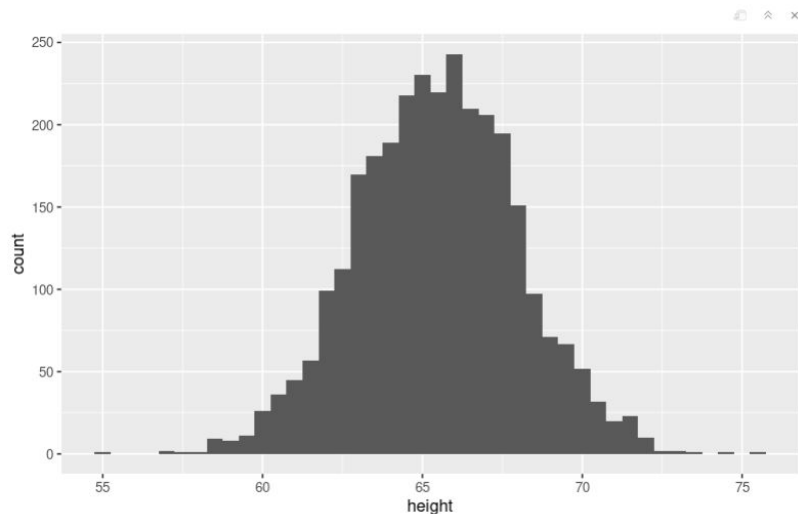
Put your graph of the 100 confidence intervals for your 100 samples here.



How many of your 100 confidence intervals actually contained the population mean? 84

Group 2

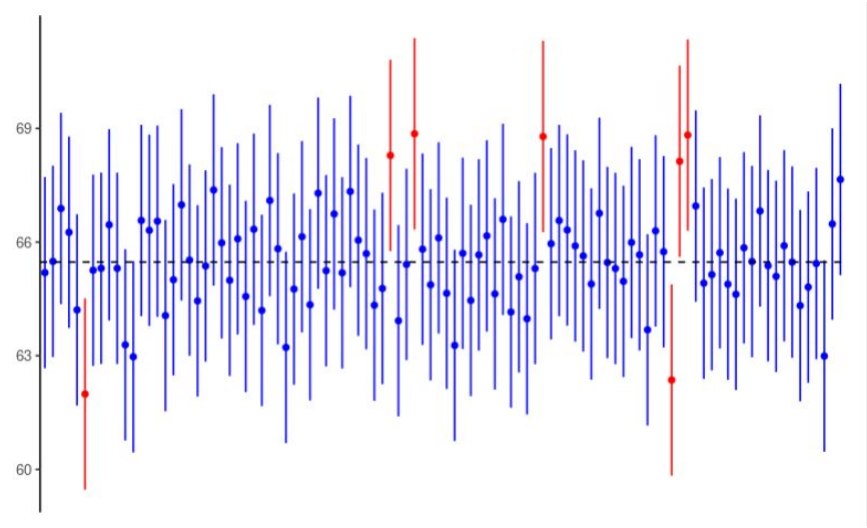
Put your graph showing the distribution of heights in the population here.



Describe the population distribution of heights, following our standard procedure (center, spread, shape): the shape of the histogram is a bell, the histogram has a center at 65 and a relatively tight spread.

How many of the 100 95% confidence intervals (calculated using the actual standard error) do you predict will contain the population mean height? I expect there to be 95 out of the 100 intervals that contain the mean, this is because the meaning of the 95% confidence interval means that over a large sample size approximately 95% will contain the mean

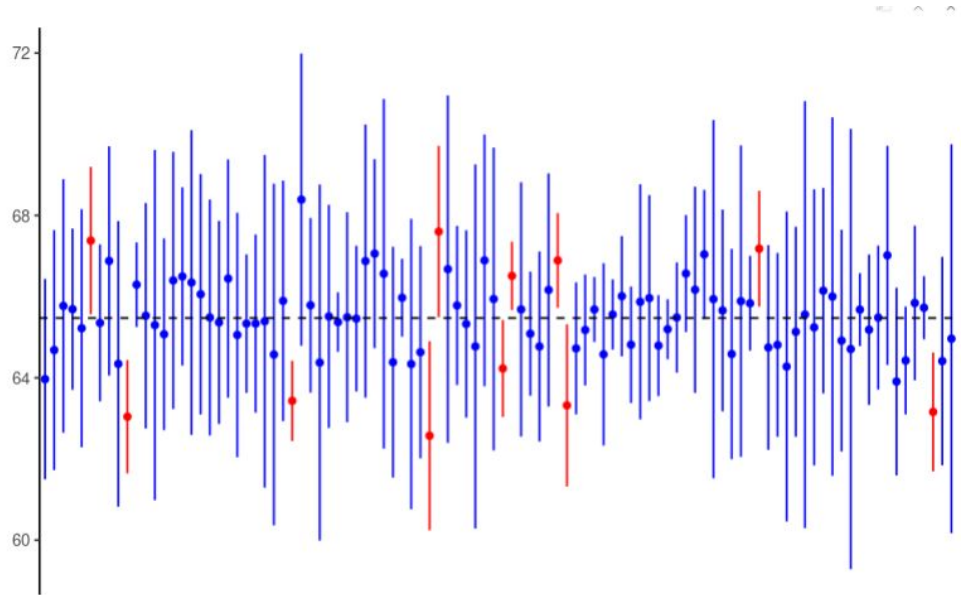
Put your graph of the 100 confidence intervals for your 100 samples here.



How many of your 100 confidence intervals actually contained the population mean? 93 did

How many of the 100 95% confidence intervals (calculated using the **estimated** standard error) do you predict will contain the population mean height? I predict that it will be 95% of the samples again

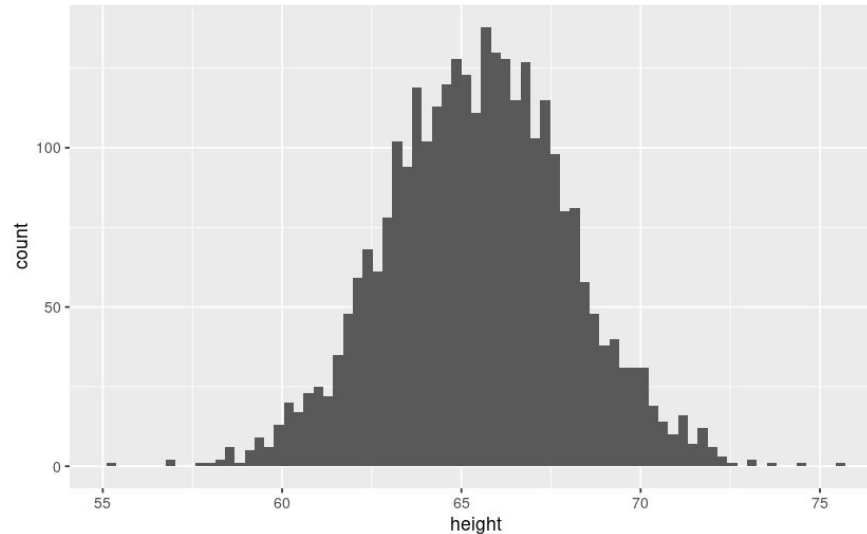
Put your graph of the 100 confidence intervals for your 100 samples here.



How many of your 100 confidence intervals actually contained the population mean? 89 of the 100 contained the median

Group 3

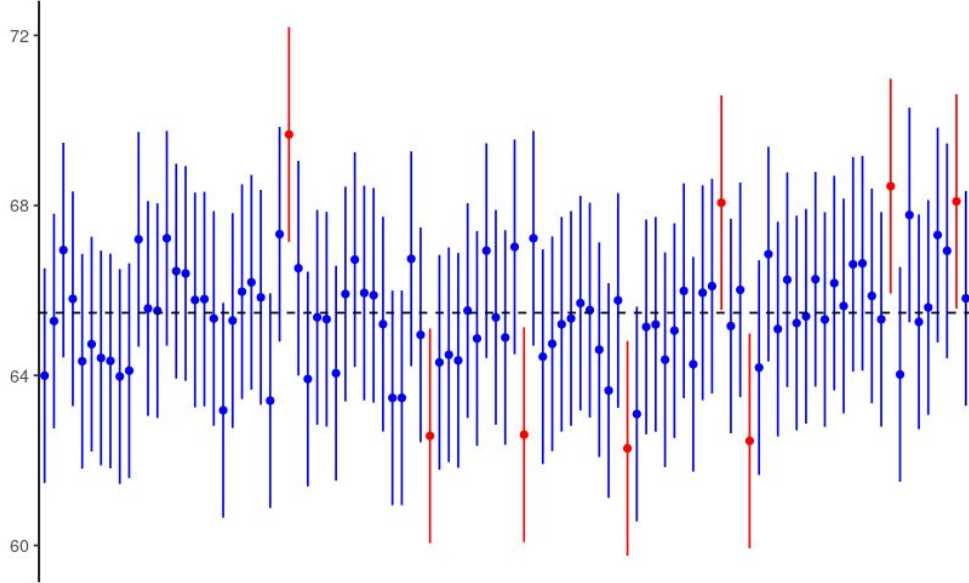
Put your graph showing the distribution of heights in the population here.



Describe the population distribution of heights, following our standard procedure (center, spread, shape):
The population distribution has a bell shape centered around 65.48 with a standard deviation of 2.53.

How many of the 100 95% confidence intervals (calculated using the actual standard error) do you predict will contain the population mean height? 95

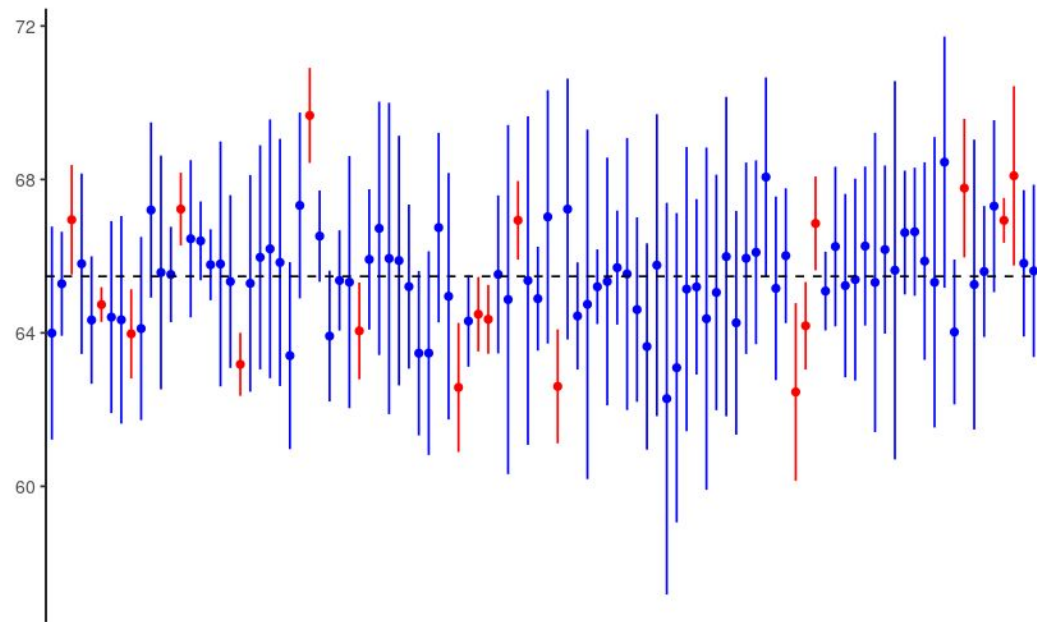
Put your graph of the 100 confidence intervals for your 100 samples here.



How many of your 100 confidence intervals actually contained the population mean? 92

How many of the 100 95% confidence intervals (calculated using the **estimated** standard error) do you predict will contain the population mean height? 95

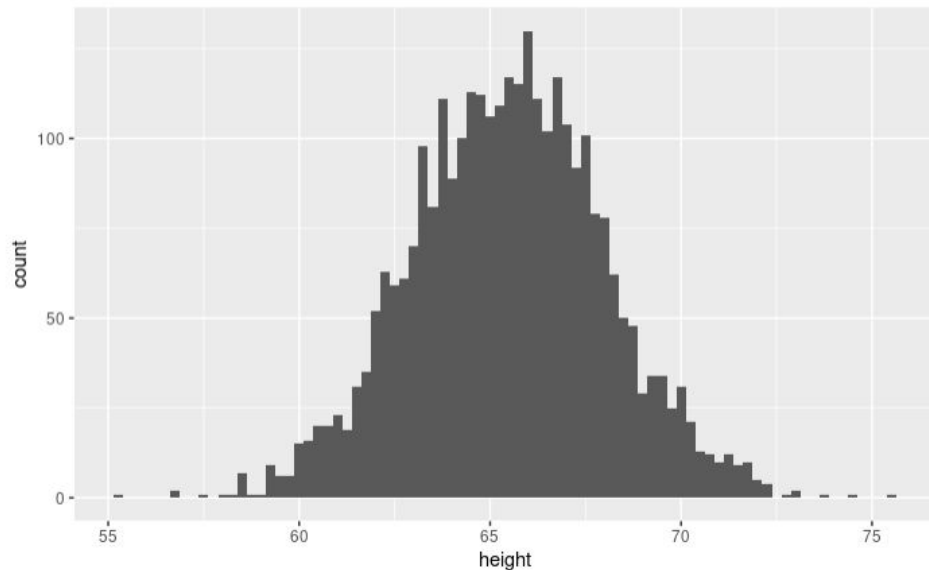
Put your graph of the 100 confidence intervals for your 100 samples here.



How many of your 100 confidence intervals actually contained the population mean? 83

Group 4

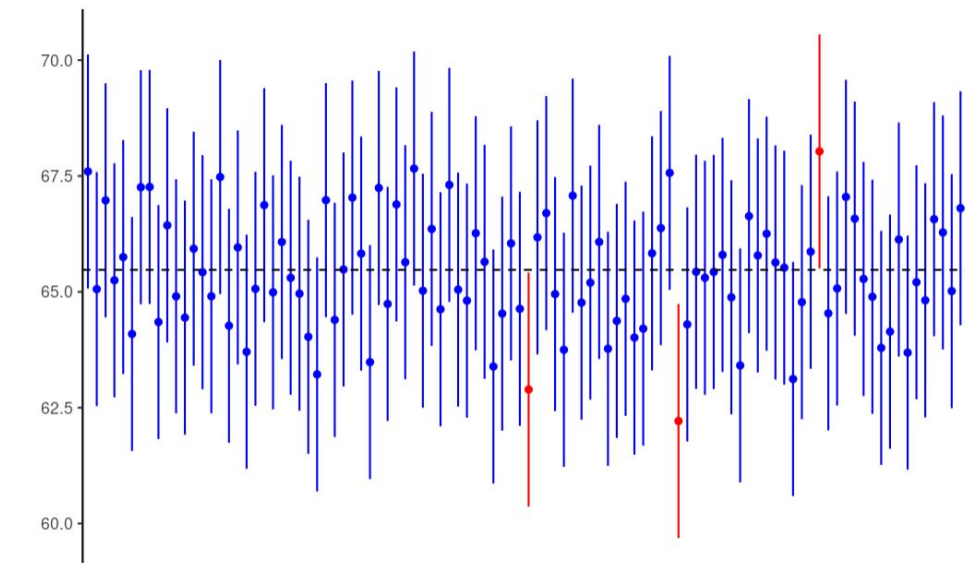
Put your graph showing the distribution of heights in the population here.



Describe the population distribution of heights, following our standard procedure (center, spread, shape):
The population distribution is bell shaped. The population has a mean height of 65.47385. The data ranges from 60.53076 to 70.45626.

How many of the 100 95% confidence intervals (calculated using the actual standard error) do you predict will contain the population mean height?

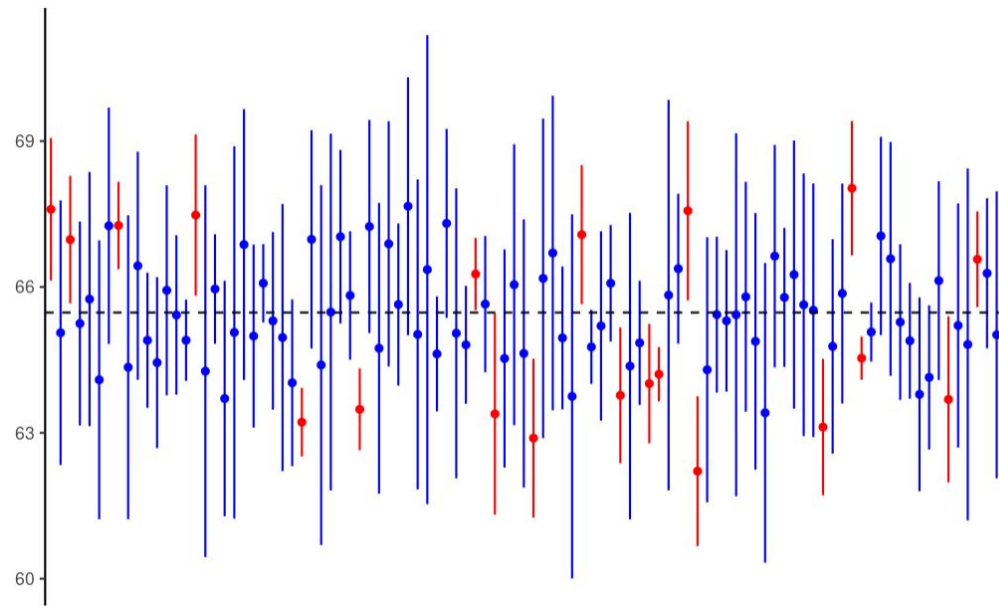
Put your graph of the 100 confidence intervals for your 100 samples here.



How many of your 100 confidence intervals actually contained the population mean? 97 intervals

How many of the 100 95% confidence intervals (calculated using the **estimated** standard error) do you predict will contain the population mean height?

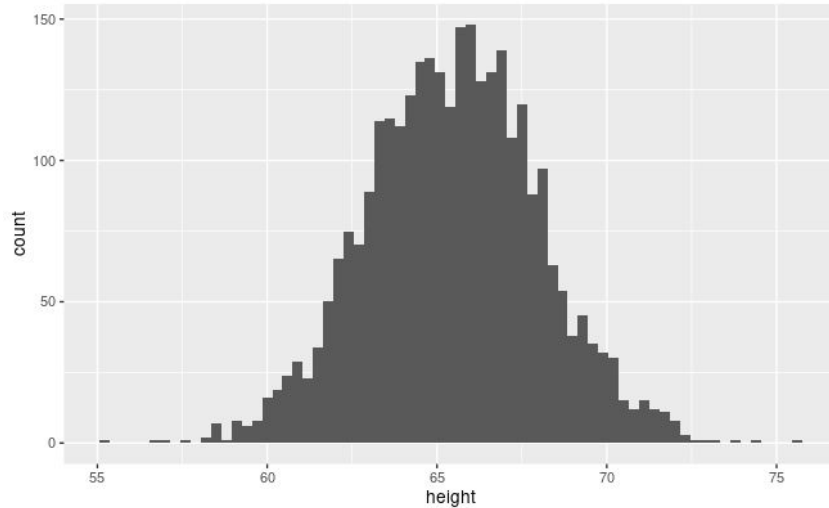
Put your graph of the 100 confidence intervals for your 100 samples here.



How many of your 100 confidence intervals actually contained the population mean? 79 intervals

Group 5

Put your graph showing the distribution of heights in the population here.

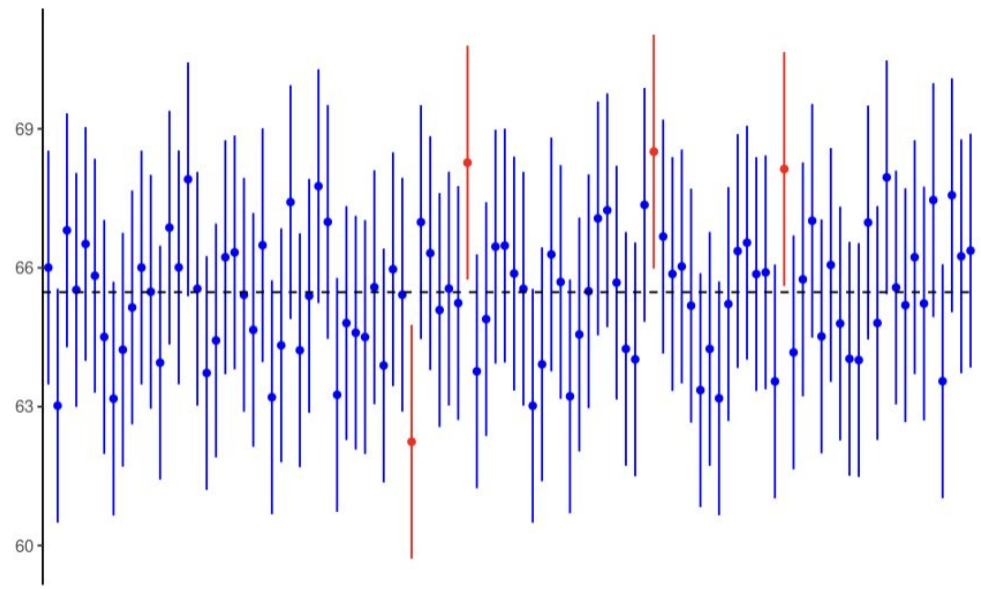


Describe the population distribution of heights, following our standard procedure (center, spread, shape):

We have a normal bell shaped distribution with the mean height at 65.47" and roughly 68% of all the people within our sample are within 2.53" of the average. 95% of everyone within our sample has a height between 63.73" and 65.51".

How many of the 100 95% confidence intervals (calculated using the actual standard error) do you predict will contain the population mean height? 95%

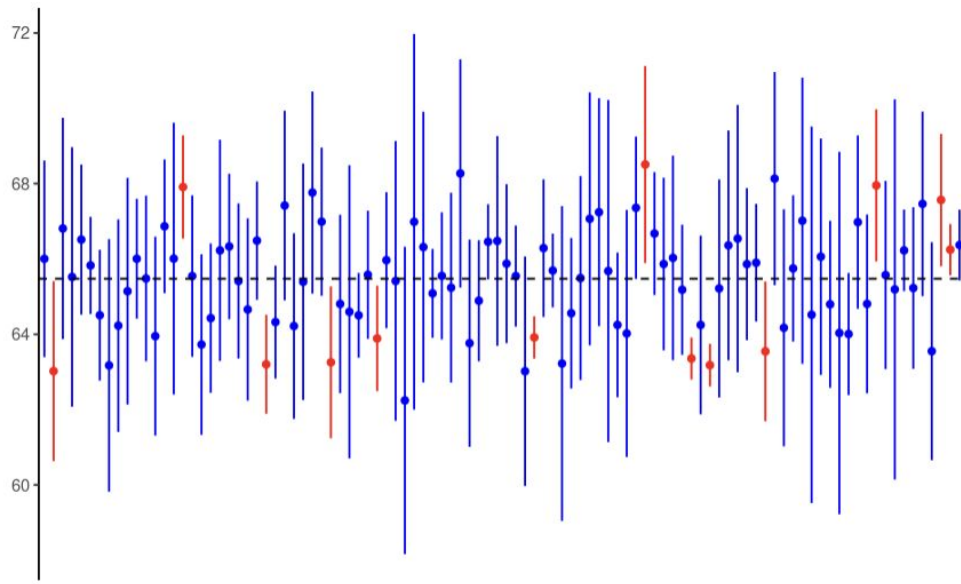
Put your graph of the 100 confidence intervals for your 100 samples here.



How many of your 100 confidence intervals actually contained the population mean? 96

How many of the 100 95% confidence intervals (calculated using the **estimated** standard error) do you predict will contain the population mean height? 95%

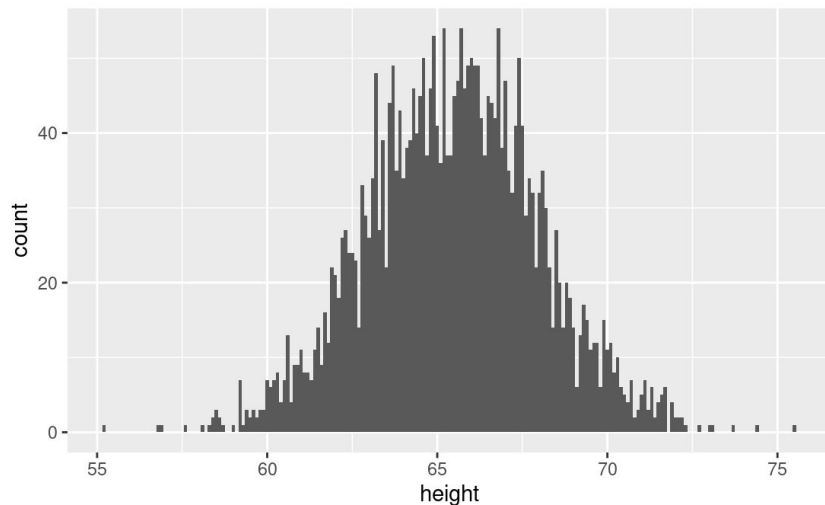
Put your graph of the 100 confidence intervals for your 100 samples here.



How many of your 100 confidence intervals actually contained the population mean? 87

Group 6

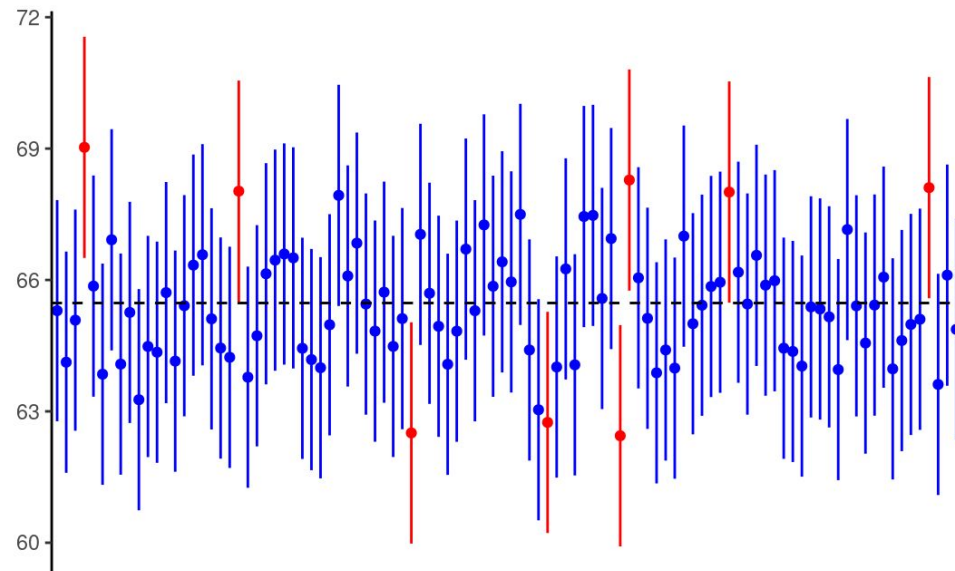
Put your graph showing the distribution of heights in the population here.



Describe the population distribution of heights, following our standard procedure (center, spread, shape): this graph appears to take on a more bell curve sort of shape, suggesting no particular favor in one height direction. Most points tend to center around the height of 65.47385, with variations of 2.525765 between points.

How many of the 100 95% confidence intervals (calculated using the actual standard error) do you predict will contain the population mean height? Um, going off of a random guess, I think maybe 75 is a good number.

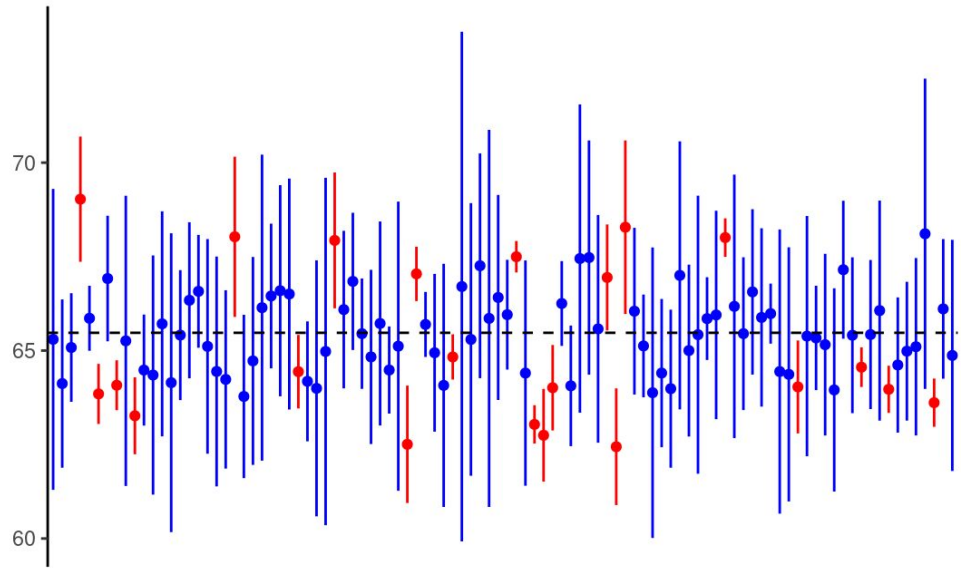
Put your graph of the 100 confidence intervals for your 100 samples here.



How many of your 100 confidence intervals actually contained the population mean? 92

How many of the 100 95% confidence intervals (calculated using the **estimated** standard error) do you predict will contain the population mean height? Um, probably a lower number like 35 or something

Put your graph of the 100 confidence intervals for your 100 samples here.



How many of your 100 confidence intervals actually contained the population mean? 78

Group 7

Put your graph showing the distribution of heights in the population here.

Describe the population distribution of heights, following our standard procedure (center, spread, shape):

How many of the 100 95% confidence intervals (calculated using the actual standard error) do you predict will contain the population mean height?

Put your graph of the 100 confidence intervals for your 100 samples here.

How many of your 100 confidence intervals actually contained the population mean?

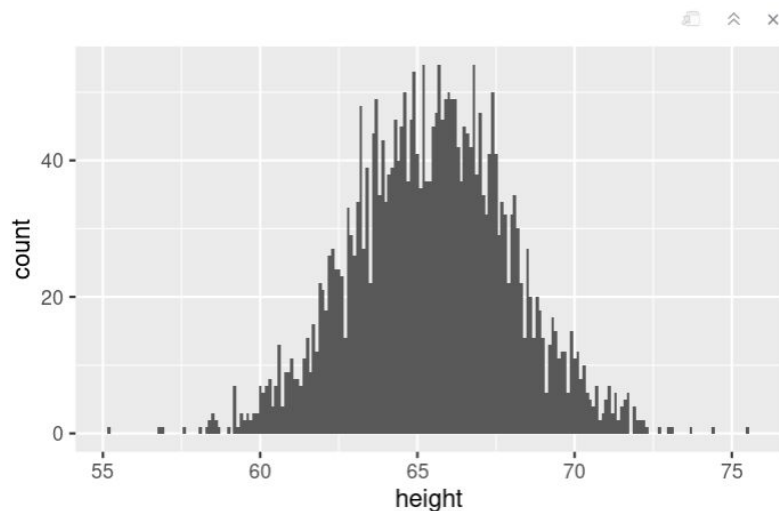
How many of the 100 95% confidence intervals (calculated using the **estimated** standard error) do you predict will contain the population mean height?

Put your graph of the 100 confidence intervals for your 100 samples here.

How many of your 100 confidence intervals actually contained the population mean?

Group 8

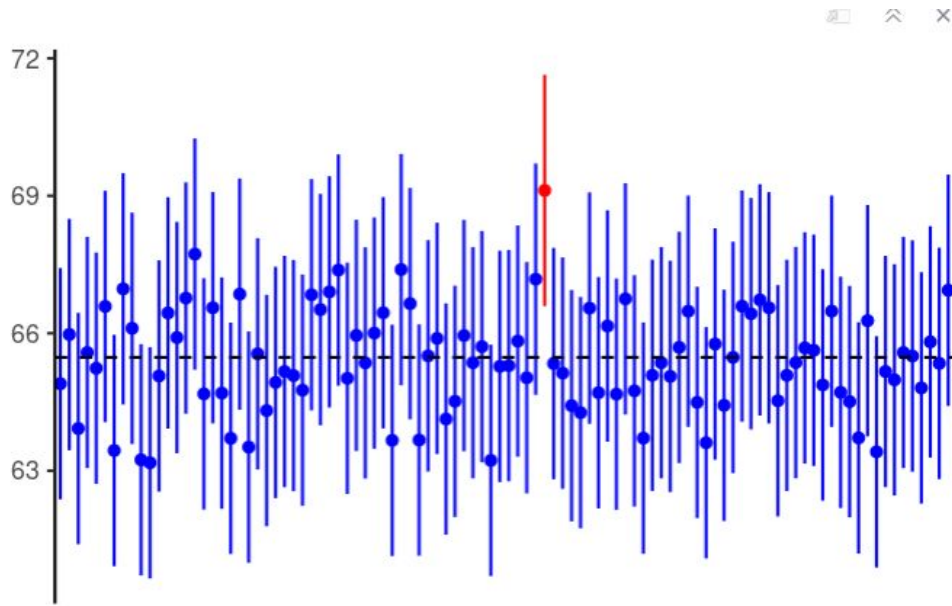
Put your graph showing the distribution of heights in the population here.



Describe the population distribution of heights, following our standard procedure (center, spread, shape):
The data shows the distributions of heights in a population. The average heights in this data of the population is 65.5 inches tall. Most of the data that is on the graph is between the heights of 63.73 and 67.16. While there is some outliers at heights of 55 inches and above 75 inches. The graph has the shape of a bell shaped graph, with all data points surrounding the mean.

How many of the 100 95% confidence intervals (calculated using the actual standard error) do you predict will contain the population mean height?

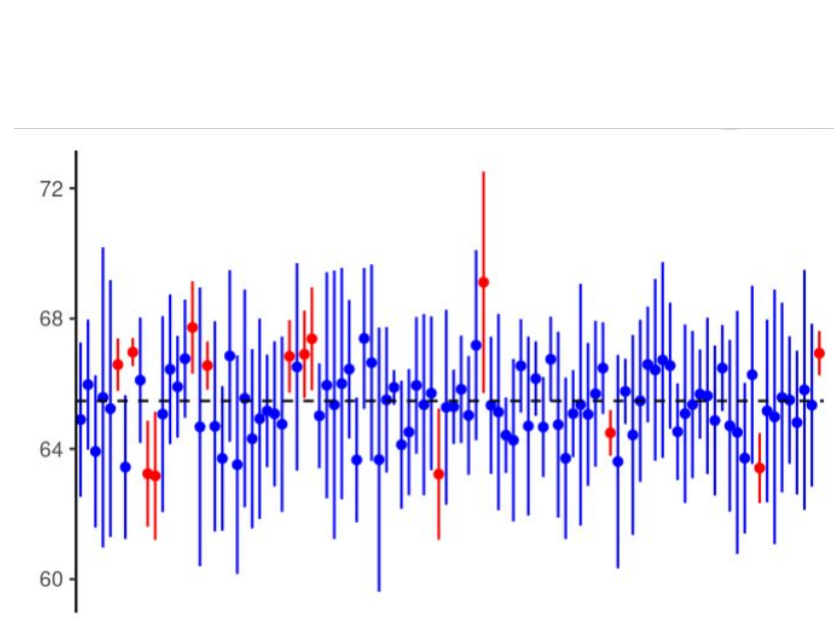
Put your graph of the 100 confidence intervals for your 100 samples here.



How many of your 100 confidence intervals actually contained the population mean? 99 intervals

How many of the 100 95% confidence intervals (calculated using the **estimated** standard error) do you predict will contain the population mean height?

Put your graph of the 100 confidence intervals for your 100 samples here.



How many of your 100 confidence intervals actually contained the population mean? 86 intervals

Group 9

Put your graph showing the distribution of heights in the population here.

Describe the population distribution of heights, following our standard procedure (center, spread, shape):

How many of the 100 95% confidence intervals (calculated using the actual standard error) do you predict will contain the population mean height?

Put your graph of the 100 confidence intervals for your 100 samples here.

How many of your 100 confidence intervals actually contained the population mean?

How many of the 100 95% confidence intervals (calculated using the **estimated** standard error) do you predict will contain the population mean height?

Put your graph of the 100 confidence intervals for your 100 samples here.

How many of your 100 confidence intervals actually contained the population mean?

Group 10

Put your graph showing the distribution of heights in the population here.

Describe the population distribution of heights, following our standard procedure (center, spread, shape):

How many of the 100 95% confidence intervals (calculated using the actual standard error) do you predict will contain the population mean height?

Put your graph of the 100 confidence intervals for your 100 samples here.

How many of your 100 confidence intervals actually contained the population mean?

How many of the 100 95% confidence intervals (calculated using the **estimated** standard error) do you predict will contain the population mean height?

Put your graph of the 100 confidence intervals for your 100 samples here.

How many of your 100 confidence intervals actually contained the population mean?